Process Creation Investigation – arp.exe

Objective

The objective of this investigation is to verify that arp.exe process execution events are successfully collected by Elastic SIEM using Sysmon logs. This investigation demonstrates how Address Resolution Protocol (ARP) activity can be monitored and analyzed to identify legitimate network administration activity as well as potential reconnaissance performed by an attacker.

Command Executed

```
arp -a
```

Detection Method (KQL)

`process.name : "ARP.EXE"`
If your environment indexes the command differently, use the query that successfully returns the event, such as `process.args : "arp"`.

Evidence Collected

Timestamp: 2026-07-20T16:38:22.621Z
Process Name: ARP.EXE
Host Name: Sepisecurity
User Domain: Sepisecurity
Process ID (PID): 24152
Event Code: 1 (Sysmon Process Creation)

Analysis

The investigation confirmed that arp.exe executed successfully on the monitored Windows endpoint. The event identified the execution time, user, host, and process identifier. No suspicious parent process, unusual command-line arguments, or indicators of malicious behavior were observed. The activity is consistent with legitimate network administration used to view the local ARP cache.

Security Impact

The arp command is commonly used by system administrators to troubleshoot network connectivity. However, attackers also use ARP during the reconnaissance phase to identify neighboring devices, map local networks, and gather information before lateral movement. Monitoring arp.exe execution helps SOC analysts detect potentially suspicious network discovery activity when correlated with other security events.

Conclusion

This investigation demonstrates the ability to detect and analyze Windows process creation events using Elastic SIEM and Sysmon. The collected evidence indicates legitimate administrative activity, and no indicators of compromise were identified. The investigation highlights practical skills in KQL searching, endpoint telemetry analysis, and professional SOC documentation.